

**Engineering Protection in  
Multi-Mode Superconducting Circuits**

by

**Eli J. Weissler**

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Committee Members:

Josh Combes, Chair

András Gyenis, Co-Chair

Scott Glancy

Florent Lecocq

Daniel Slichter

Joe Aumentado

Weissler, Eli J. (Ph.D., Electrical Engineering)

Engineering Protection in Multi-Mode Superconducting Circuits

Thesis directed by Prof. Josh Combes

This is my single-page abstract!

## Dedication

This is my single-page dedication!

## Acknowledgements

This is my single-page acknowledgements section!

## Contents

### Chapter

|          |                                                                     |          |
|----------|---------------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                                 | <b>1</b> |
| 1.1      | Quantum Computing . . . . .                                         | 2        |
| 1.2      | Circuit Quantum Electrodynamics (cQED or Circuit QED) . . . . .     | 2        |
| 1.2.1    | Superconductivity . . . . .                                         | 2        |
| 1.2.2    | LC Resonator . . . . .                                              | 2        |
| 1.2.3    | Transmission Lines . . . . .                                        | 2        |
| 1.2.4    | Basic Circuit Quantization . . . . .                                | 2        |
| 1.3      | Common Superconducting Qubits and Operations . . . . .              | 2        |
| 1.3.1    | Cooper Pair Box & Transmon . . . . .                                | 2        |
| 1.3.2    | Fluxonium . . . . .                                                 | 2        |
| 1.3.3    | Single-Qubit Gates . . . . .                                        | 2        |
| 1.3.4    | Two-Qubit Gates . . . . .                                           | 2        |
| 1.3.5    | Dispersive Readout . . . . .                                        | 2        |
| 1.4      | Numerical Modeling of Circuit QED Systems . . . . .                 | 2        |
| 1.4.1    | Finite Difference & Spectral Methods . . . . .                      | 3        |
| 1.4.2    | Multi-Mode & Composite Systems . . . . .                            | 3        |
| 1.5      | Protecting Quantum Information from Noise . . . . .                 | 3        |
| 1.5.1    | Noise Filtering vs. Error Protection vs. Error Correction . . . . . | 3        |
| 1.5.2    | Protected Qubits and Estimating Lifetimes . . . . .                 | 3        |

|          |                                                    |           |
|----------|----------------------------------------------------|-----------|
| 1.5.3    | Quantum Error Correction (QEC)                     | 3         |
| 1.5.4    | Noise Filtering in Circuit QED                     | 3         |
| <b>2</b> | Automated Mode Separation and Circuit Quantization | <b>4</b>  |
| <b>3</b> | Matrix Free Methods for Composite Systems          | <b>5</b>  |
| <b>4</b> | Circuit Enumeration                                | <b>6</b>  |
| <b>5</b> | Two-Qubit Gates for the Soft $0 - \pi$ Qubit       | <b>7</b>  |
| <b>6</b> | $\cos M\varphi$ Qudits and Rotor Codes             | <b>8</b>  |
|          | <b>Glossary</b>                                    | <b>8</b>  |
|          | <b>References</b>                                  | <b>9</b>  |
|          | <b>Appendix</b>                                    |           |
| <b>A</b> | Fun animal facts                                   | <b>10</b> |
| <b>B</b> | Ode to Spot                                        | <b>12</b> |

## Tables

### Table

## Figures

### Figure

|     |                                               |    |
|-----|-----------------------------------------------|----|
| 1.1 | Test figure for this chapter. . . . .         | 1  |
| 1.2 | Test circuit figure for this chapter. . . . . | 1  |
| 2.1 | Test figure for this chapter. . . . .         | 4  |
| 2.2 | Test circuit figure for this chapter. . . . . | 4  |
| 3.1 | Test figure for this chapter. . . . .         | 5  |
| 3.2 | Test circuit figure for this chapter. . . . . | 5  |
| 4.1 | Test figure for this chapter. . . . .         | 6  |
| 4.2 | Test circuit figure for this chapter. . . . . | 6  |
| 5.1 | Test figure for this chapter. . . . .         | 7  |
| 5.2 | Test circuit figure for this chapter. . . . . | 7  |
| 6.1 | Test figure for this chapter. . . . .         | 8  |
| 6.2 | Test circuit figure for this chapter. . . . . | 8  |
| A.1 | Test figure for this chapter. . . . .         | 10 |
| A.2 | Test circuit figure for this chapter. . . . . | 10 |
| B.1 | Test figure for this chapter. . . . .         | 12 |
| B.2 | Test circuit figure for this chapter. . . . . | 12 |

# Chapter 1

## Introduction

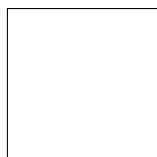


Figure 1.1: Test figure for this chapter.

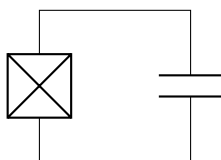


Figure 1.2: Test circuit figure for this chapter.

People love quantum computing I'm focused on an aspect of it.

Background is stuff that I don't want to explain to people in the later chapters. Superconducting basics. Dressed basis two qubit gates, stabilizers for error correcting codes, basic circuit quantization using the devoret recipe.

Explaining dispersive readout, voltage control on transmon, flux tuning, etc. The basics. Transmission lines.

Discuss protection vs. error correction.

Noise estimates.

Testing my bibliography here [1].

Testing approximation for circuit quantum electrodynamics (cQED) and quantum error correction (QEC)

## **1.1 Quantum Computing**

## **1.2 Circuit Quantum Electrodynamics (cQED or Circuit QED)**

### **1.2.1 Superconductivity**

### **1.2.2 LC Resonator**

### **1.2.3 Transmission Lines**

### **1.2.4 Basic Circuit Quantization**

## **1.3 Common Superconducting Qubits and Operations**

### **1.3.1 Cooper Pair Box & Transmon**

### **1.3.2 Fluxonium**

### **1.3.3 Single-Qubit Gates**

### **1.3.4 Two-Qubit Gates**

### **1.3.5 Dispersive Readout**

## **1.4 Numerical Modeling of Circuit QED Systems**

Numerical methods are a powerful tool for understanding the behavior of circuit QED systems. There are a limited number of analytical solutions to the Hamiltonians that describe cQED systems. Numerical methods are, to be clear, approximate solutions. However with modest computational resources, one can prove invaluable in system design and verifying analytical expressions. Numerical methods are also necessary for simulating time dynamics, where even fewer analytical solutions exist.

### 1.4.1 Finite Difference & Spectral Methods

To model a system numerically, we must first define an appropriate basis. Unsurprisingly, this differs based on the circuit. In Superconducting circuits there are two distinct Hilbert spaces that emerge. The first is a standard continuous position (phase), momentum (charge) conjugate pair, as in the simple harmonic oscillator. The second is a  $2\pi$  periodic compact phase, or position, coordinate and integer valued charge, or momentum, coordinate. This would represent e.g., one unit cell of a Transmon potential.

To model the continuous Hilbert, we must identify a basis that spans possible eigenfunctions  $\psi(x)$ , which will all be square normalizable,  $\int_{-\infty}^{\infty} dx |\psi(x)|^2 = 1$ . The two most common choices are first a discrete position or momentum eigenstates

$$\hat{n}|n\rangle = n|n\rangle, \quad \hat{\phi}|n\rangle = i\frac{\partial}{\partial n}|n\rangle \quad \& \quad \hat{n}|\phi\rangle = -i\frac{\partial}{\partial \phi}|\phi\rangle, \quad \hat{\phi}|\phi\rangle = \phi|\phi\rangle, \quad (1.1)$$

which are sampled at a finite number of points. Another straightforward choice is the harmonic oscillator basis

$$|m\rangle \quad (1.2)$$

### 1.4.2 Multi-Mode & Composite Systems

## 1.5 Protecting Quantum Information from Noise

### 1.5.1 Noise Filtering vs. Error Protection vs. Error Correction

### 1.5.2 Protected Qubits and Estimating Lifetimes

### 1.5.3 Quantum Error Correction (QEC)

### 1.5.4 Noise Filtering in Circuit QED

## Chapter 2

### Automated Mode Separation and Circuit Quantization

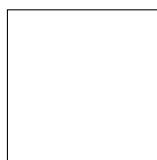


Figure 2.1: Test figure for this chapter.

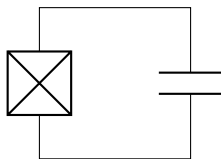


Figure 2.2: Test circuit figure for this chapter.

## Chapter 3

### Matrix Free Methods for Composite Systems

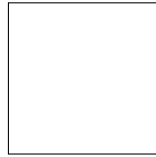


Figure 3.1: Test figure for this chapter.

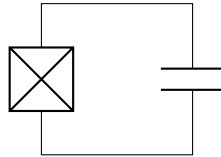


Figure 3.2: Test circuit figure for this chapter.

## Chapter 4

### Circuit Enumeration

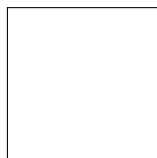


Figure 4.1: Test figure for this chapter.

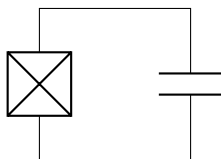


Figure 4.2: Test circuit figure for this chapter.

## Chapter 5

### Two-Qubit Gates for the Soft $0 - \pi$ Qubit

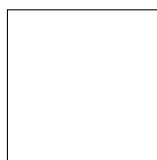


Figure 5.1: Test figure for this chapter.

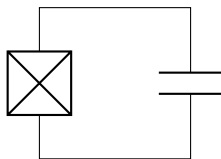


Figure 5.2: Test circuit figure for this chapter.

## Chapter 6

### $\cos M\varphi$ Qudits and Rotor Codes

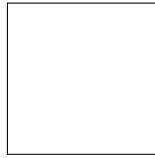


Figure 6.1: Test figure for this chapter.

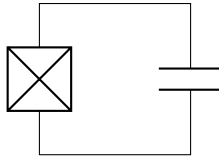


Figure 6.2: Test circuit figure for this chapter.

## Glossary

**cQED** circuit quantum electrodynamics

**QEC** quantum error correction

## References

- [1] F. Arute, K. Arya, R. Babbush, D. Bacon, J. C. Bardin, R. Barends, R. Biswas, S. Boixo, F. G. S. L. Brandao, D. A. Buell, B. Burkett, Y. Chen, Z. Chen, B. Chiaro, R. Collins, W. Courtney, A. Dunsworth, E. Farhi, B. Foxen, A. Fowler, C. Gidney, M. Giustina, R. Graff, K. Guerin, S. Habegger, M. P. Harrigan, M. J. Hartmann, A. Ho, M. Hoffmann, T. Huang, T. S. Humble, S. V. Isakov, E. Jeffrey, Z. Jiang, D. Kafri, K. Kechedzhi, J. Kelly, P. V. Klimov, S. Knysh, A. Korotkov, F. Kostritsa, D. Landhuis, M. Lindmark, E. Lucero, D. Lyakh, S. Mandrà, J. R. McClean, M. McEwen, A. Megrant, X. Mi, K. Michielsen, M. Mohseni, J. Mutus, O. Naaman, M. Neeley, C. Neill, M. Y. Niu, E. Ostby, A. Petukhov, J. C. Platt, C. Quintana, E. G. Rieffel, P. Roushan, N. C. Rubin, D. Sank, K. J. Satzinger, V. Smelyanskiy, K. J. Sung, M. D. Trevithick, A. Vainsencher, B. Villalonga, T. White, Z. J. Yao, P. Yeh, A. Zalcman, H. Neven, and J. M. Martinis, “Quantum supremacy using a programmable superconducting processor”, *Nature* **574**, 505 (2019) (Cited on p. 1).

## Appendix A

### Fun animal facts

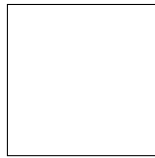


Figure A.1: Test figure for this chapter.

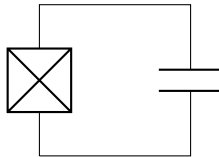


Figure A.2: Test circuit figure for this chapter.

**About appendices:** Each appendix follow the same page-numbering rules as a regular chapter; the first page of a (multi-page) appendix is not numbered. Here are some facts about dogs!

- (1) Dalmatians are born without spots! They are born with plain white coats with their first spots appearing after they are 1week old.
- (2) Dogs sweat through their foot pads to help keep them cool. They also keep cool by panting.
- (3) Greyhounds are the world's fastest dogs with the ability to reach up to 45 mph.
- (4) Every dog has a unique nose print with no two alike.

- (5) The Basenji, an African wolf dog, does not bark in a normal way but may yodel or scream when excited!
- (6) Snoopy, from Charles M. Schultz's "Peanuts" comic strip, is a beagle.
- (7) A dog's sense of smell is 1000 times greater than a human!
- (8) Nine percent of dog owners will have a birthday party for their pet.
- (9) Dogs have 28 baby teeth and 42 permanent teeth.

And here are some other animal facts:

- (1) Fleas can jump 350 times its body length.
- (2) Hummingbirds are the only birds that can fly backwards.
- (3) Crocodiles cannot stick their tongue out.
- (4) Starfish do not have a brain.
- (5) Slugs have 4 noses.
- (6) Only female mosquitoes bite.
- (7) Polar bear skin is black!
- (8) The only mammal capable of flight is the bat.
- (9) A newborn kangaroo is the size of a lima bean.

I took these facts from [animalmedical.org](http://animalmedical.org), which was the first Google search result for "list of fun facts about animals".

## Appendix B

### Ode to Spot



Figure B.1: Test figure for this chapter.

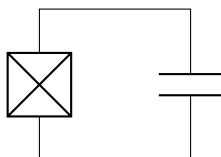


Figure B.2: Test circuit figure for this chapter.

**(Data, Stardate 1403827)** (A one-page chapter — page must be numbered!) Throughout the ages, from Keats to Giorchamo, poets have composed “odes” to individuals who have had a profound effect upon their lives. In keeping with that tradition I have written my next poem . . . in honor of my cat. I call it. . . Ode. . . to Spot. (Shot of Geordi and Worf in audience, looking mystified at each other.)

Felus cattus, is your taxonomic nomenclature  
 an endothermic quadruped, carnivorous by nature?  
 Your visual, olfactory, and auditory senses  
 contribute to your hunting skills, and natural defenses.  
 I find myself intrigued by your sub-vocal oscillations,

a singular development of cat communications  
that obviates your basic hedonistic predilection  
for a rhythmic stroking of your fur to demonstrate affection.  
A tail is quite essential for your acrobatic talents;  
you would not be so agile if you lacked its counterbalance.  
And when not being utilized to aid in locomotion,  
It often serves to illustrate the state of your emotion.

(Commander Riker begins to applaud, until a glance from Counselor Troi brings him to a halt.)

Commander Riker, you have anticipated my denouement. However, the sentiment is appreciated. I  
will continue.

O Spot, the complex levels of behavior you display  
connote a fairly well-developed cognitive array.  
And though you are not sentient, Spot, and do not comprehend  
I nonetheless consider you a true and valued friend.